

Solution Requirements

Project: FitBuddy – AI Fitness Plan Generator Using Gemini

Purpose: Project-track submission material adapted to the supplied AI/ML/Gen-AI template.

Functional requirements

- Collect user fitness profile and goal information.
- Generate a personalized seven-day workout plan using Gemini.
- Display the generated plan in the web interface.
- Persist user/plan data through the database layer.
- Accept feedback to revise a plan.
- Provide AI-generated nutrition tips.
- Provide an admin-oriented view if supported by the current implementation.

Non-functional requirements

- Keep API keys out of source control.
- Validate user input and handle upstream failures.
- Use readable, maintainable modules and templates.
- Present clear UI feedback and reasonable responsive behavior.
- Document setup, configuration, and known limitations.

Constraints and assumptions

Generation depends on a valid Gemini API key, service availability, and model access. Fitness outputs are general information and should not replace professional medical advice.